

Differential Expression Part 2

- Reminder: Assignment #2 is due today at 11:59pm
- The next assignment is a 1-page summary of your proposed project (see next), due on March 3rd
- Today's lab activity is 06_DE_edgeR.qmd

edgeR Analysis Pipeline Overview

- 1) Load count data (column 1 = gene names, subsequent columns are counts for samples, row names are headings)
- 2) Create “groups” for your samples that you want to compare (e.g., group 1 = WT unstressed, group 2 = WT EtOH stressed, group 3 = mutant unstressed, group 4 = mutant stressed)
- 3) Create a “design matrix” that assigns groups to the samples in your count table
- 4) Create a “contrast matrix” for your comparisons of interest (e.g., group 1 (WT unstressed) vs group 2 (WT EtOH stressed))

edgeR Analysis Pipeline Overview

- 5) Filter out genes with low counts
- 6) Normalize to total reads per sample and composition (TMM)
- 7) Estimate dispersion (negative binomial)
- 8) Test for differential expression via quasi-likelihood F test
- 9) Write table of outputs (logFC, p-values, FDR-adjusted p-values)

edgeR TREAT (Test for Differential Expression Relative to a Threshold)

BIOINFORMATICS

ORIGINAL PAPER

Vol. 25 no. 6 2009, pages 765–771
doi:10.1093/bioinformatics/btp053

Gene expression

Testing significance relative to a fold-change threshold is a TREAT

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The idea is that post-hoc application of a logFC cutoff after obtaining FDR-corrected p-values, then the FDR correction may be wrong (because FDR was applied to the whole list and not a subset)

Applying FDR correction to p-values only generated after applying a FC cutoff solves this problem